

Weyl and Dirac Semimetals in Three-Dimensional Solids

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Preface

This document presents my personal notes on Weyl and Dirac semimetals in three-dimensional solids, focusing on their band structure, topological properties, and the emergence of relativistic quasiparticles in condensed matter systems. The notes are written as part of my ongoing effort to understand how linear band crossings near the Fermi level give rise to Weyl and Dirac fermion-like excitations in solids.

The study also includes discussions on key concepts such as Berry curvature, chiral anomaly, surface Fermi arcs, and symmetry protection mechanisms that stabilize these phases. These entries reflect my gradual understanding of topological semimetals and their role in modern condensed matter physics.

This section is based on a review paper that I have read as part of my literature survey on Weyl and Dirac semimetals. It is not original work, but a set of personal notes and understanding developed while studying the following article:

Reference Paper

Weyl and Dirac semimetals in three-dimensional solids

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Weyl and Dirac semimetals are three-dimensional phases of matter with gapless electronic excitations that are protected by topology and symmetry. As three-dimensional analogs of graphene, they have generated significant recent interest. Deep connections exist with particle physics models of relativistic chiral fermions and, despite their gaplessness, with solid-state topological and Chern insulators.

Their characteristic electronic properties lead to protected surface states and novel responses to applied electric and magnetic fields. The theoretical foundations of these phases, their proposed realizations in solid-state systems, and recent experiments on candidate materials, as well as their relation to other states of matter, are reviewed in this work.

Chapter 1

Accidental Degeneracies

A general 2×2 Hamiltonian can be written as a linear combination of the three Pauli matrices:

$$H = f_1\sigma_1 + f_2\sigma_2 + f_3\sigma_3, \quad (1.1)$$

where f_1, f_2, f_3 are real coefficients.

The *von Neumann–Wigner theorem* addresses the natural question of degeneracies: under what conditions on the parameters f_1, f_2, f_3 do the energy eigenvalues become degenerate? Equivalently, when does the energy difference

$$\Delta E = 0 \quad (1.2)$$

occur?

For the above Hamiltonian, the eigenvalues can be obtained using the characteristic equation:

$$\det(H - \lambda I) = 0. \quad (1.3)$$

Substituting the Hamiltonian,

$$\det \begin{pmatrix} f_3 - \lambda & f_1 - if_2 \\ f_1 + if_2 & -f_3 - \lambda \end{pmatrix} = 0. \quad (1.4)$$

Solving for λ , we obtain the eigenvalues:

$$E_{\pm} = \pm \sqrt{f_1^2 + f_2^2 + f_3^2}. \quad (1.5)$$

Therefore, the energy splitting is

$$\Delta E = E_+ - E_-, \quad (1.6)$$

which gives

$$\Delta E = 2\sqrt{f_1^2 + f_2^2 + f_3^2}. \quad (1.7)$$

Hence, the only way to obtain degeneracy,

$$\Delta E = 0, \quad (1.8)$$

is to require

$$f_1 = f_2 = f_3 = 0. \quad (1.9)$$

Thus, in the absence of any additional symmetry constraints, all three parameters must be tuned simultaneously to achieve degeneracy.

However, when *time-reversal symmetry* is imposed, the situation changes. In certain cases, the Hamiltonian is constrained such that the coefficient of σ_y vanishes automatically. Then the Hamiltonian effectively depends only on two parameters:

$$H = f_1\sigma_1 + f_3\sigma_3. \tag{1.10}$$

The degeneracy condition then reduces to

$$f_1 = f_3 = 0. \tag{1.11}$$

Therefore, under time-reversal symmetry, only two parameters need to be tuned to obtain degeneracy.

The precise reason why the σ_y term disappears under time-reversal symmetry will be discussed later.

Chapter 2

The Dirac Equation

2.1 Lorentz Invariance

A brief note on Lorentz invariance:

Lorentz invariance is one of the fundamental symmetry principles of special relativity. It states that the laws of physics remain unchanged under Lorentz transformations between different inertial frames of reference. In particular, the spacetime interval in Minkowski spacetime,

$$s^2 = t^2 - x^2 - y^2 - z^2,$$

remains invariant under such transformations.

To better understand the significance of Lorentz invariance, it is useful to compare it with the non-relativistic Galilean transformation. We will examine how the spacetime interval behaves under a general Galilean transformation and show that, unlike Lorentz transformations, the Minkowski spacetime interval is not preserved.

Let's consider a general case :

The most general Galilean transformation between two inertial frames can be written as

$$x' = x - V_x t, \tag{2.1}$$

$$y' = y - V_y t, \tag{2.2}$$

$$z' = z - V_z t, \tag{2.3}$$

$$t' = t. \tag{2.4}$$

In vector form, this may be written compactly as

$$\vec{r}' = R(\vec{r} - \vec{V}t), \tag{2.5}$$

where

$$\vec{r} = (x, y, z), \quad \vec{r}' = (x', y', z'),$$

and R is a spatial rotation matrix.

Expanding Eq. (2.5),

$$\vec{r}' = R\vec{r} - R\vec{V}t. \tag{2.6}$$

To study the spacetime interval, we compute the squared magnitude:

$$|\vec{r}'|^2 = (R(\vec{r} - \vec{V}t)) \cdot (R(\vec{r} - \vec{V}t)). \quad (2.7)$$

Using the matrix representation of the dot product,

$$\vec{a} \cdot \vec{b} = \vec{a}^T \vec{b},$$

we obtain

$$|\vec{r}'|^2 = (\vec{r} - \vec{V}t)^T R^T R (\vec{r} - \vec{V}t). \quad (2.8)$$

For a rotation matrix,

$$R^T R = I, \quad (2.9)$$

where I is the identity matrix.

This follows from the fact that rotations preserve Euclidean lengths and angles. For any vector $\vec{a}$,

$$|R\vec{a}|^2 = (R\vec{a})^T (R\vec{a}) = \vec{a}^T R^T R \vec{a}.$$

Since rotations must preserve lengths,

$$|R\vec{a}|^2 = |\vec{a}|^2 = \vec{a}^T \vec{a},$$

which implies

$$R^T R = I.$$

Substituting Eq. (2.9) into Eq. (2.8),

$$|\vec{r}'|^2 = (\vec{r} - \vec{V}t)^T (\vec{r} - \vec{V}t). \quad (2.10)$$

Expanding,

$$|\vec{r}'|^2 = |\vec{r}|^2 - 2t(\vec{V} \cdot \vec{r}) + V^2 t^2. \quad (2.11)$$

The Minkowski spacetime interval is defined as

$$s^2 = t^2 - x^2 - y^2 - z^2 = t^2 - |\vec{r}|^2. \quad (2.12)$$

Similarly, in the transformed frame,

$$s'^2 = t'^2 - |\vec{r}'|^2. \quad (2.13)$$

Since $t' = t$, substituting Eq. (2.11) gives

$$s'^2 = t^2 - \left(|\vec{r}|^2 - 2t(\vec{V} \cdot \vec{r}) + V^2 t^2 \right) \quad (2.14)$$

$$= t^2 - |\vec{r}|^2 + 2t(\vec{V} \cdot \vec{r}) - V^2 t^2. \quad (2.15)$$

Using Eq. (2.12),

$$s'^2 = s^2 + 2t(\vec{V} \cdot \vec{r}) - V^2 t^2. \quad (2.16)$$

Therefore,

$$s'^2 \neq s^2, \quad (2.17)$$

for a general nonzero velocity $\vec{V}$.

Hence, the Minkowski spacetime interval is not preserved under Galilean transformations. This shows that Galilean transformations are not Lorentz invariant.

Now we can see how the spacetime interval is preserved under the Lorentz transformation thus by showing that it is lorentz invariant.

We now repeat the same analysis for Lorentz transformations and examine how the Minkowski spacetime interval behaves under relativistic transformations between inertial frames.

Consider two inertial frames S and S' , where S' moves with a constant velocity $\vec{V}$ relative to S . For simplicity, let the boost be along the x -direction, so that

$$\vec{V} = (v, 0, 0).$$

The Lorentz transformation is then given by

$$x' = \gamma(x - vt), \quad (2.18)$$

$$y' = y, \quad (2.19)$$

$$z' = z, \quad (2.20)$$

$$t' = \gamma(t - vx), \quad (2.21)$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2}}. \quad (2.22)$$

In vector notation,

$$\vec{r} = (x, y, z), \quad \vec{r}' = (x', y', z').$$

The Minkowski spacetime interval is defined as

$$s^2 = t^2 - |\vec{r}|^2, \quad (2.23)$$

where

$$|\vec{r}|^2 = x^2 + y^2 + z^2.$$

Similarly, in the transformed frame,

$$s'^2 = t'^2 - |\vec{r}'|^2. \quad (2.24)$$

Substituting the Lorentz transformations,

$$s'^2 = \gamma^2(t - vx)^2 - \gamma^2(x - vt)^2 - y^2 - z^2. \quad (2.25)$$

Expanding both squares,

$$s'^2 = \gamma^2(t^2 - 2vxt + v^2x^2) - \gamma^2(x^2 - 2vxt + v^2t^2) - y^2 - z^2. \quad (2.26)$$

Distributing the minus sign,

$$s'^2 = \gamma^2(t^2 - 2vxt + v^2x^2 - x^2 + 2vxt - v^2t^2) - y^2 - z^2. \quad (2.27)$$

The mixed terms cancel:

$$-2vxt + 2vxt = 0.$$

Thus,

$$s'^2 = \gamma^2 [t^2(1 - v^2) - x^2(1 - v^2)] - y^2 - z^2. \quad (2.28)$$

Factoring out $(1 - v^2)$,

$$s'^2 = \gamma^2(1 - v^2)(t^2 - x^2) - y^2 - z^2. \quad (2.29)$$

Using Eq. (2.22),

$$\gamma^2 = \frac{1}{1 - v^2},$$

we obtain

$$\gamma^2(1 - v^2) = 1. \quad (2.30)$$

Therefore,

$$s'^2 = t^2 - x^2 - y^2 - z^2. \quad (2.31)$$

Using Eq. (2.23),

$$s'^2 = s^2. \quad (2.32)$$

Thus, unlike the Galilean transformation, the Lorentz transformation preserves the Minkowski spacetime interval. This invariance forms the geometric foundation of special relativity and ensures that all inertial observers agree on the causal structure of spacetime.

2.2 Schrodinger Equation and Minkowski Metric

Considering the Schrödinger equation in operator notation,

$$i\partial_t\psi = -\frac{1}{2m}\nabla^2\psi, \quad (2.33)$$

we now examine how the equation changes under a Lorentz transformation.

Since space and time coordinates mix under relativistic transformations, we first study how the derivatives transform. In particular, we begin with the time derivative appearing on the left-hand side:

$$i\partial_t\psi.$$

To determine how $\partial_t\psi$ transforms, we use the multivariable chain rule.

We consider the wavefunction as

$$\psi = \psi(x', t'),$$

where

$$x' = x'(x, t), \quad t' = t'(x, t).$$

For simplicity, we consider a Lorentz boost only along the x -direction.

Before applying this to the Lorentz transformation, let us first derive the general multivariable chain rule.

Suppose

$$f = f(x', y'),$$

where

$$x' = x'(x, y), \quad y' = y'(x, y).$$

The total differential of f is

$$df = \frac{\partial f}{\partial x'} dx' + \frac{\partial f}{\partial y'} dy'. \quad (2.34)$$

Now expand dx' and dy' :

$$dx' = \frac{\partial x'}{\partial x} dx + \frac{\partial x'}{\partial y} dy, \quad (2.35)$$

$$dy' = \frac{\partial y'}{\partial x} dx + \frac{\partial y'}{\partial y} dy. \quad (2.36)$$

Substituting into the expression for df ,

$$\begin{aligned} df &= \frac{\partial f}{\partial x'} \left(\frac{\partial x'}{\partial x} dx + \frac{\partial x'}{\partial y} dy \right) \\ &\quad + \frac{\partial f}{\partial y'} \left(\frac{\partial y'}{\partial x} dx + \frac{\partial y'}{\partial y} dy \right). \end{aligned} \quad (2.37)$$

Expanding,

$$\begin{aligned} df &= \left(\frac{\partial f}{\partial x'} \frac{\partial x'}{\partial x} + \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial x} \right) dx \\ &\quad + \left(\frac{\partial f}{\partial x'} \frac{\partial x'}{\partial y} + \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial y} \right) dy. \end{aligned} \quad (2.38)$$

But from ordinary multivariable calculus,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy. \quad (2.39)$$

Comparing coefficients of dx and dy , we obtain

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial x'} \frac{\partial x'}{\partial x} + \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial x}, \quad (2.40)$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial x'} \frac{\partial x'}{\partial y} + \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial y}. \quad (2.41)$$

Therefore, in operator form,

$$\partial_x = \frac{\partial x'}{\partial x} \partial_{x'} + \frac{\partial y'}{\partial x} \partial_{y'}. \quad (2.42)$$

Applying this idea to Lorentz transformations,

$$\partial_t = \frac{\partial x'}{\partial t} \partial_{x'} + \frac{\partial t'}{\partial t} \partial_{t'}. \quad (2.43)$$

We now complete the derivation of how partial derivatives transform under a Lorentz boost.

Starting from the chain rule,

$$\partial_t = \frac{\partial x'}{\partial t} \partial_{x'} + \frac{\partial t'}{\partial t} \partial_{t'}, \quad (2.44)$$

and using the Lorentz transformation

$$x' = \gamma(x - vt), \quad t' = \gamma(t - vx), \quad (2.45)$$

we compute the required partial derivatives.

First,

$$\frac{\partial x'}{\partial t} = \frac{\partial}{\partial t}[\gamma(x - vt)] = -\gamma v, \quad (2.46)$$

and

$$\frac{\partial t'}{\partial t} = \frac{\partial}{\partial t}[\gamma(t - vx)] = \gamma. \quad (2.47)$$

Substituting into the chain rule gives

$$\partial_t = (-\gamma v) \partial_{x'} + \gamma \partial_{t'}. \quad (2.48)$$

Factoring out γ , we obtain

$$\partial_t = \gamma(\partial_{t'} - v \partial_{x'}). \quad (2.49)$$

Similarly, for the spatial derivative,

$$\partial_x = \frac{\partial x'}{\partial x} \partial_{x'} + \frac{\partial t'}{\partial x} \partial_{t'}. \quad (2.50)$$

We compute

$$\frac{\partial x'}{\partial x} = \gamma, \quad \frac{\partial t'}{\partial x} = -\gamma v. \quad (2.51)$$

Thus,

$$\partial_x = \gamma \partial_{x'} - \gamma v \partial_{t'}, \quad (2.52)$$

which can be written as

$$\partial_x = \gamma(\partial_{x'} - v \partial_{t'}). \quad (2.53)$$

Therefore, the final transformation rules are

$$\partial_t = \gamma(\partial_{t'} - v \partial_{x'}), \quad (2.54)$$

$$\partial_x = \gamma(\partial_{x'} - v \partial_{t'}). \quad (2.55)$$

We now show explicitly that the Schrödinger equation is not invariant under Lorentz transformations by transforming both sides of the equation.

We begin with the Lorentz transformation between two inertial frames S and S' , where S' moves with velocity v along the x -axis:

$$x' = \gamma(x - vt), \quad t' = \gamma(t - vx), \quad (2.56)$$

with

$$\gamma = \frac{1}{\sqrt{1 - v^2}}. \quad (2.57)$$

Using the chain rule, the derivatives transform as

$$\partial_t = \gamma (\partial_{t'} - v \partial_{x'}) , \quad (2.58)$$

$$\partial_x = \gamma (\partial_{x'} - v \partial_{t'}) . \quad (2.59)$$

The one-dimensional Schrödinger equation is

$$i \partial_t \psi(x, t) = -\frac{1}{2m} \partial_x^2 \psi(x, t). \quad (2.60)$$

Substituting the transformed time derivative gives

$$\begin{aligned} i \partial_t \psi &= i \gamma (\partial_{t'} - v \partial_{x'}) \psi \\ &= i \gamma \partial_{t'} \psi - i \gamma v \partial_{x'} \psi. \end{aligned} \quad (2.61)$$

For the spatial derivative,

$$\begin{aligned} \partial_x^2 &= \gamma^2 (\partial_{x'} - v \partial_{t'})^2 \\ &= \gamma^2 (\partial_{x'}^2 - 2v \partial_{x'} \partial_{t'} + v^2 \partial_{t'}^2) . \end{aligned} \quad (2.62)$$

Therefore,

$$-\frac{1}{2m} \partial_x^2 \psi = -\frac{\gamma^2}{2m} (\partial_{x'}^2 - 2v \partial_{x'} \partial_{t'} + v^2 \partial_{t'}^2) \psi. \quad (2.63)$$

The transformed Schrödinger equation becomes

$$i \gamma (\partial_{t'} - v \partial_{x'}) \psi = -\frac{\gamma^2}{2m} (\partial_{x'}^2 - 2v \partial_{x'} \partial_{t'} + v^2 \partial_{t'}^2) \psi. \quad (2.64)$$

The transformed Schrödinger equation does not preserve its original form under Lorentz transformations; it acquires mixed space-time derivative terms and second-order time derivatives, demonstrating that the Schrödinger equation is not Lorentz invariant.

2.3 Klein–Gordon Equation

Since the Schrödinger equation is not Lorentz invariant, it is not compatible with the requirements of special relativity. In particular, the equation treats time and space derivatives asymmetrically: it is first order in time but second order in space. A relativistic wave equation should place space and time on a more symmetric footing and remain consistent with Lorentz transformations.

To construct such an equation, we begin from the relativistic energy–momentum relation,

$$E^2 = p^2 c^2 + m^2 c^4. \quad (2.65)$$

For convenience, we adopt *natural units*,

$$\hbar = c = 1, \quad (2.66)$$

under which the fundamental constants are absorbed into the choice of units. This is permissible because physical laws are independent of the unit system used; setting $c = 1$ measures time in units of length, while $\hbar = 1$ expresses energy, momentum, inverse length, and inverse time in the same units. The relativistic dispersion relation then simplifies to

$$E^2 = p^2 + m^2. \quad (2.67)$$

To obtain a quantum wave equation, energy and momentum are promoted to operators acting on a wavefunction $\psi(x, t)$. Using the standard quantum mechanical correspondence,

$$E \rightarrow i\partial_t, \quad (2.68)$$

$$p \rightarrow -i\nabla, \quad (2.69)$$

we substitute these operator forms into the relativistic energy relation:

$$E^2\psi = (p^2 + m^2)\psi. \quad (2.70)$$

Applying the operator substitutions gives

$$(i\partial_t)^2\psi = [(-i\nabla)^2 + m^2]\psi. \quad (2.71)$$

Evaluating the operators,

$$(i\partial_t)^2\psi = -\partial_t^2\psi, \quad (2.72)$$

$$(-i\nabla)^2\psi = -\nabla^2\psi. \quad (2.73)$$

Therefore,

$$-\partial_t^2\psi = (-\nabla^2 + m^2)\psi. \quad (2.74)$$

Rearranging terms yields

$$\boxed{(\partial_t^2 - \nabla^2 + m^2)\psi = 0} \quad (2.75)$$

which is the Klein–Gordon equation in natural units.

Restoring c and $\hbar$, the equation becomes

$$\boxed{\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2 c^2}{\hbar^2}\right)\psi = 0} \quad (2.76)$$

This equation is second order in both time and space derivatives and is consistent with the relativistic energy momentum relation, making it Lorentz invariant.

Why Not Quantize the Unsquared Relativistic Relation?

One may ask why the relativistic energy relation is not quantized directly using

$$E = \sqrt{p^2 + m^2}. \quad (2.77)$$

Applying the standard operator substitutions,

$$E \rightarrow i\partial_t, \quad (2.78)$$

$$p \rightarrow -i\nabla, \quad (2.79)$$

leads formally to

$$i\partial_t\psi = \sqrt{-\nabla^2 + m^2} \psi. \quad (2.80)$$

The difficulty arises from the square root of the differential operator. Factoring out the mass term gives

$$\hat{H} = m\sqrt{1 - \frac{\nabla^2}{m^2}}. \quad (2.81)$$

Using the binomial expansion,

$$(1 + x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \dots, \quad (2.82)$$

with

$$x = -\frac{\nabla^2}{m^2}, \quad (2.83)$$

the Hamiltonian becomes

$$\hat{H} = m - \frac{1}{2m}\nabla^2 - \frac{1}{8m^3}\nabla^4 + \frac{1}{16m^5}\nabla^6 + \dots \quad (2.84)$$

Rather than obtaining a finite differential equation, one obtains an infinite series of increasingly higher-order derivative operators.

A finite-order differential operator is local: its action at a point depends only on the wavefunction and finitely many of its derivatives in an infinitesimal neighbourhood of that point. By contrast, an infinite-order differential operator is generally nonlocal, since its action involves infinitely many derivatives of the wavefunction.

Such nonlocal operators are generally undesirable in relativistic quantum mechanics, where one seeks local wave equations with a well-defined spacetime structure. To avoid this difficulty, Klein and Gordon instead began from the squared relativistic relation,

$$E^2 = p^2 + m^2, \quad (2.85)$$

which leads to the local second-order Klein–Gordon equation derived above.

2.4 Problems With Klein–Gordon

Second Order Differential Equation

Consider the first-order differential equation with the initial condition given equation

$$y(0) = 1.$$

$$\frac{dy}{dt} = y,$$

$$\frac{dy}{y} = dt,$$

$$\ln y = t + C,$$

$$y(t) = Ce^t. \tag{2.86}$$

Applying the initial condition fixes the constant,

$$y(0) = Ce^0 = 1 \Rightarrow C = 1,$$

$$y(t) = e^t. \tag{2.87}$$

The evolution is fully determined once the value of the function at a single time is specified. Now consider the second-order differential equation with the initial condition.

$$\frac{d^2y}{dt^2} = -y,$$

$$y(0) = 1.$$

The general solution is written as

$$y(t) = A \cos t + B \sin t.$$

Imposing the initial condition gives

$$y(0) = A \cos 0 + B \sin 0 = A,$$

which fixes

$$A = 1.$$

The solution therefore reduces to

$$y(t) = \cos t + B \sin t. \tag{2.88}$$

A first-order differential equation requires one initial condition to determine a unique solution, whereas a second-order equation requires two.

The Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2) \psi = 0,$$

contains a second-order time derivative. Knowledge of the wavefunction alone,

$$\psi(\mathbf{x}, 0),$$

is insufficient to determine its future evolution. One must also specify

$$\partial_t \psi(\mathbf{x}, 0).$$

The evolution is therefore determined by the pair

$$(\psi, \partial_t \psi), \tag{2.89}$$

rather than by the wavefunction alone, unlike the Schrödinger equation.

The Klein–Gordon equation (refer to Eq. (3.76)) is second order in time. Therefore, its evolution is not determined by $\psi(\mathbf{x}, 0)$ alone. To uniquely determine the time evolution, one must also specify the initial time derivative $\partial_t \psi(\mathbf{x}, 0)$.

To uniquely determine the solution, one must also specify

$$\partial_t \psi(\mathbf{x}, 0).$$

Hence, the "state" of the system is effectively given by the pair

$$(\psi, \partial_t \psi),$$

rather than the wavefunction alone, unlike in the Schrödinger equation.

Probability density problem

The Klein–Gordon equation (see Eq. (3.76)) admits both positive and negative energy solutions. This can be seen by considering plane wave solutions of the form

$$\psi(\mathbf{x}, t) = e^{-i(Et - \mathbf{p} \cdot \mathbf{x})}. \tag{2.90}$$

Substituting this into Eq. (3.76), gives the relativistic dispersion relation

$$E^2 = \mathbf{p}^2 + m^2.$$

Thus the energy spectrum splits into two branches,

$$E = \pm \sqrt{\mathbf{p}^2 + m^2}. \tag{2.91}$$

This presence of both positive and negative energy solutions plays a crucial role in the structure of the conserved current.

The conserved current associated with Eq. (3.76) is given by

$$j^\mu = \frac{i}{2m} (\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*). \tag{2.92}$$

The time component of this current defines the density,

$$\rho = j^0 = \frac{i}{2m} (\psi^* \partial_t \psi - \psi \partial_t \psi^*). \quad (2.93)$$

For a plane wave solution, Eq. (3.90) we have,

$$\partial_t \psi = -iE\psi. \quad (2.94)$$

Substituting into the expression for the density,

$$\begin{aligned} \rho &= \frac{i}{2m} [\psi^* (-iE\psi) - \psi (iE\psi^*)] \\ &= \frac{E}{m} |\psi|^2. \end{aligned}$$

This shows that the sign of the density is directly determined by the sign of the energy,

$$\rho \propto E. \quad (2.95)$$

Therefore, positive energy solutions correspond to positive density, while negative energy solutions lead to negative density.

Unlike non-relativistic quantum mechanics, where the probability density is always positive definite,

$$\rho = |\psi|^2 \geq 0, \quad (2.96)$$

the Klein–Gordon density is not positive definite.

This shows that the quantity ρ cannot be interpreted as a probability density.

2.5 The Dirac's Equation

The Klein Gordon equation resolved the incompatibility between the Schrödinger equation and special relativity by incorporating the relativistic energy–momentum relation. However, as discussed earlier, it introduced new difficulties: the equation is second order in time and its conserved density is not positive definite.

Dirac sought an alternative relativistic wave equation that would remain first order in both space and time derivatives, more closely resembling the structure of the Schrödinger equation.

Starting from the relativistic dispersion relation,

$$E^2 = p^2 + m^2, \quad (2.97)$$

Dirac proposed that the Hamiltonian should be *linear* in both momentum and mass. He therefore assumed the form

$$H = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m, \quad (2.98)$$

where

$$\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3),$$

and α_i and β are quantities yet to be determined.

Since Eq. (2.97) requires

$$H^2 = p^2 + m^2, \quad (2.99)$$

we now square the assumed Hamiltonian and compare the result with the relativistic energy relation in order to determine the allowed form of α_i and β .

Squaring Eq. (2.98),

$$\begin{aligned} H^2 &= (\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m)^2 \\ &= (\boldsymbol{\alpha} \cdot \mathbf{p})^2 + (\boldsymbol{\alpha} \cdot \mathbf{p})\beta m + \beta m(\boldsymbol{\alpha} \cdot \mathbf{p}) + \beta^2 m^2. \end{aligned} \quad (2.100)$$

Writing the momentum term explicitly,

$$\boldsymbol{\alpha} \cdot \mathbf{p} = \alpha_i p_i, \quad (2.101)$$

where summation over repeated indices is implied.

$$\begin{aligned} (\boldsymbol{\alpha} \cdot \mathbf{p})^2 &= (\alpha_i p_i)(\alpha_j p_j) \\ &= \alpha_i \alpha_j p_i p_j. \end{aligned} \quad (2.102)$$

$$(\boldsymbol{\alpha} \cdot \mathbf{p})\beta m + \beta m(\boldsymbol{\alpha} \cdot \mathbf{p}) = (\alpha_i \beta + \beta \alpha_i) p_i m. \quad (2.103)$$

$$H^2 = \alpha_i \alpha_j p_i p_j + (\alpha_i \beta + \beta \alpha_i) p_i m + \beta^2 m^2. \quad (2.104)$$

Comparing Eq. (2.104) with the required relativistic form,

$$H^2 = p^2 + m^2, \quad (2.105)$$

Comparing Eq. (2.104) with Eq. (2.99), we require the matrices α_i and β to satisfy certain algebraic constraints.

The mixed terms must vanish:

$$\alpha_i \beta + \beta \alpha_i = 0. \quad (2.106)$$

The mass term must reproduce m^2 :

$$\beta^2 = I. \quad (2.107)$$

The momentum term must reproduce p^2 . Therefore,

$$\alpha_i^2 = I, \quad (2.108)$$

and for distinct spatial indices,

$$\alpha_i \alpha_j + \alpha_j \alpha_i = 0, \quad i \neq j. \quad (2.109)$$

These conditions may be written compactly using the anti-commutator notation,

$$\{A, B\} \equiv AB + BA. \quad (2.110)$$

The algebra then becomes

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij}I, \quad (2.111)$$

$$\{\alpha_i, \beta\} = 0, \quad (2.112)$$

$$\beta^2 = I. \quad (2.113)$$

Thus, the problem reduces to finding objects α_i and β satisfying these algebraic relations.

Dirac found matrices obeying this algebra. We shall not discuss here how these matrices were originally discovered. Instead, we simply introduce their standard representation and verify explicitly that they satisfy all constraints obtained from the assumed form of the Hamiltonian.

The standard Dirac representation of β is

$$\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad (2.114)$$

where I denotes the 2×2 identity matrix.

We first verify Eq. (2.107).

Squaring Eq. (2.114),

$$\begin{aligned} \beta^2 &= \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \\ &= \begin{pmatrix} I^2 & 0 \\ 0 & (-I)^2 \end{pmatrix} \\ &= \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \\ &= I_4. \end{aligned} \quad (2.115)$$

Hence,

$$\beta^2 = I_4, \quad (2.116)$$

as required.

The standard form of the α_i matrices is

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad (2.117)$$

where σ_i are the Pauli matrices.

We now verify the required properties.

First, squaring Eq. (2.117),

$$\begin{aligned} \alpha_i^2 &= \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \\ &= \begin{pmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_i^2 \end{pmatrix}. \end{aligned} \quad (2.118)$$

Using the Pauli matrix identity

$$\sigma_i^2 = I, \quad (2.119)$$

we obtain

$$\alpha_i^2 = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} = I_4. \quad (2.120)$$

Thus Eq. (2.108) is satisfied.

Next, consider the anti-commutation relation between distinct α matrices.

For $i \neq j$,

$$\begin{aligned} \alpha_i \alpha_j &= \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} \\ &= \begin{pmatrix} \sigma_i \sigma_j & 0 \\ 0 & \sigma_i \sigma_j \end{pmatrix}, \end{aligned} \quad (2.121)$$

$$\alpha_j \alpha_i = \begin{pmatrix} \sigma_j \sigma_i & 0 \\ 0 & \sigma_j \sigma_i \end{pmatrix}. \quad (2.122)$$

Therefore,

$$\alpha_i \alpha_j + \alpha_j \alpha_i = \begin{pmatrix} \sigma_i \sigma_j + \sigma_j \sigma_i & 0 \\ 0 & \sigma_i \sigma_j + \sigma_j \sigma_i \end{pmatrix}. \quad (2.123)$$

Using the Pauli matrix algebra,

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I, \quad (2.124)$$

we obtain

$$\alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij}I_4. \quad (2.125)$$

Thus Eq. (2.111) is satisfied.

Finally, we verify the mixed anti-commutation relation between α_i and β .

Using Eqs. (2.114) and (2.117),

$$\begin{aligned} \alpha_i \beta &= \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \\ &= \begin{pmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{pmatrix}, \end{aligned} \quad (2.126)$$

$$\begin{aligned} \beta \alpha_i &= \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}. \end{aligned} \quad (2.127)$$

Adding the two expressions,

$$\begin{aligned}\alpha_i\beta + \beta\alpha_i &= \begin{pmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{pmatrix} + \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \\ &= 0.\end{aligned}\tag{2.128}$$

Therefore,

$$\{\alpha_i, \beta\} = 0,\tag{2.129}$$

which verifies Eq. (2.112).

Hence, the standard Dirac matrices indeed satisfy all algebraic constraints obtained from the linearised relativistic Hamiltonian.

Now, to express the Dirac equation in a more compact and manifestly relativistic form, we introduce the γ -matrices.

We define the gamma matrices as

$$\gamma^0 \equiv \beta,\tag{2.130}$$

$$\gamma^i \equiv \beta\alpha_i, \quad i = 1, 2, 3.\tag{2.131}$$

Recall the Dirac equation in Hamiltonian form,

$$i\partial_t\psi = (\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m)\psi,\tag{2.132}$$

with

$$\mathbf{p} = -i\nabla.\tag{2.133}$$

Substituting the momentum operator,

$$i\partial_t\psi = (-i\alpha_i\partial_i + \beta m)\psi.\tag{2.134}$$

Moving all terms to the left-hand side,

$$(i\partial_t + i\alpha_i\partial_i - \beta m)\psi = 0.\tag{2.135}$$

We now multiply Eq. (2.135) by β from the left:

$$\beta(i\partial_t + i\alpha_i\partial_i - \beta m)\psi = 0.\tag{2.136}$$

$$i\beta\partial_t\psi + i\beta\alpha_i\partial_i\psi - \beta^2 m\psi = 0.\tag{2.137}$$

$$\beta^2 = I,\tag{2.138}$$

$$(i\beta\partial_t + i\beta\alpha_i\partial_i - m)\psi = 0.\tag{2.139}$$

Using the definitions of the gamma matrices from Eqs. (2.130) and (2.131),

$$(i\gamma^0\partial_t + i\gamma^i\partial_i - m)\psi = 0.\tag{2.140}$$

Introducing the compact notation

$$\partial_\mu = (\partial_t, \partial_1, \partial_2, \partial_3),\tag{2.141}$$

with the Einstein summation convention over repeated indices, Eq. (2.140) may be written as

$$\boxed{(i\gamma^\mu\partial_\mu - m)\psi = 0} \tag{2.142}$$

which is the standard covariant form of the Dirac equation.

Chapter 3

Weyl Fermions

3.1 Clifford Algebra and Gamma Matrices

To understand Weyl fermions, it is useful to first become familiar with the Clifford algebra satisfied by gamma matrices appearing in the Dirac equation.

Let us therefore begin with a simple exercise.

Suppose we propose the following matrices as gamma matrices in $1 + 1$ dimensions,

$$\gamma^0 = \sigma_z, \quad \gamma^1 = i\sigma_y, \quad (3.1)$$

where

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (3.2)$$

Verify whether these matrices satisfy the Clifford algebra relation

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I} \quad (3.3)$$

for the Minkowski metric

$$\eta^{\mu\nu} = \text{diag}(1, -1). \quad (3.4)$$

Solution :

We begin from the Clifford algebra,

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} I.} \quad (3.5)$$

Before directly substituting the matrices, let us first derive the conditions they must satisfy.

Choosing $\mu = \nu = 0$ in Eq. (3.5),

$$\gamma^0 \gamma^0 + \gamma^0 \gamma^0 = 2\eta^{00} I. \quad (3.6)$$

Since

$$\eta^{00} = 1,$$

$$2(\gamma^0)^2 = 2I, \quad (3.7)$$

$$(\gamma^0)^2 = I. \quad (3.8)$$

Next, choosing $\mu = \nu = 1$,

$$\gamma^1\gamma^1 + \gamma^1\gamma^1 = 2\eta^{11}I. \quad (3.9)$$

$$\eta^{11} = -1,$$

$$2(\gamma^1)^2 = -2I, \quad (3.10)$$

$$(\gamma^1)^2 = -I. \quad (3.11)$$

Finally, choosing $\mu \neq \nu$,

$$\gamma^0\gamma^1 + \gamma^1\gamma^0 = 2\eta^{01}I. \quad (3.12)$$

Because the Minkowski metric is diagonal,

$$\eta^{01} = 0,$$

$$\gamma^0\gamma^1 + \gamma^1\gamma^0 = 0. \quad (3.13)$$

Thus, to satisfy the Clifford algebra, our candidate matrices must obey Eqs. (3.8), (3.11), and (3.13).

We now verify these conditions explicitly.

Step 1: Checking $(\gamma^0)^2 = I$

Using

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (3.14)$$

we compute

$$(\gamma^0)^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.15)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (3.16)$$

$$= I. \quad (3.17)$$

Therefore, Eq. (3.8) is satisfied.

Step 2: Checking $(\gamma^1)^2 = -I$

First compute $i\sigma_y$,

$$\gamma^1 = i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (3.18)$$

Then,

$$(\gamma^1)^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (3.19)$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.20)$$

$$= -I. \quad (3.21)$$

Thus, Eq. (3.11) is satisfied.

Step 3: Checking the anticommutation relation

First compute

$$\gamma^0 \gamma^1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (3.22)$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (3.23)$$

Next,

$$\gamma^1 \gamma^0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.24)$$

$$= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}. \quad (3.25)$$

Adding them,

$$\gamma^0 \gamma^1 + \gamma^1 \gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \quad (3.26)$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad (3.27)$$

Therefore,

$$\boxed{\{\gamma^0, \gamma^1\} = 0.} \quad (3.28)$$

Hence, the proposed matrices

$$\boxed{\gamma^0 = \sigma_z, \quad \gamma^1 = i\sigma_y} \quad (3.29)$$

indeed satisfy the Clifford algebra in $1 + 1$ dimensions.

3.2 Chirality and Weyl Reduction in $1 + 1$ Dimensions

In $(1 + 1)$ spacetime dimensions, the Dirac equation simplifies significantly. This simplification becomes even more transparent in the massless limit. We set $m = 0$ and consider one time and one spatial coordinate.

We choose the following representation of the gamma matrices:

$$\gamma^0 = \sigma_z, \quad \gamma^1 = i\sigma_y, \quad (3.30)$$

where σ_i are the Pauli matrices.

The massless Dirac equation is given by:

$$i\gamma^\mu \partial_\mu \psi = 0.$$

$$i\gamma^0 \partial_t \psi + i\gamma^1 \partial_x \psi = 0.$$

Multiplying the Dirac equation from the left by γ^0 , we obtain:

$$\gamma^0 (i\gamma^0 \partial_t \psi + i\gamma^1 \partial_x \psi) = 0.$$

$$i\partial_t \psi + i\gamma^0 \gamma^1 \partial_x \psi = 0.$$

Introducing the momentum operator $p = -i\partial_x$, this becomes:

$$i\partial_t \psi = (\gamma^0 \gamma^1) p \psi.$$

We define the chirality operator:

$$\gamma^5 \equiv \gamma^0 \gamma^1. \quad (3.31)$$

Hence,

$$i\partial_t \psi = \gamma^5 p \psi.$$

For eigenstates of γ^5 , satisfying

$$\gamma^5 \psi_\pm = \pm \psi_\pm,$$

$$i\partial_t \psi_\pm = \pm p \psi_\pm.$$

Converting to energy eigenvalues using $\psi \sim e^{-iEt}$, we obtain the dispersion relation:

$$E = \pm p. \quad (3.32)$$

This describes right and left-moving massless chiral fermions in $(1 + 1)$ dimensions.

3.3 Chirality and Weyl Reduction in $3 + 1$ Dimensions

In $(3 + 1)$ spacetime dimensions, the Dirac equation describes four-component spinors and exhibits an internal structure associated with spin and chirality. In the massless limit, the theory

decomposes into independent chiral sectors.

The Dirac equation is

$$(i\gamma^\mu \partial_\mu - m)\psi = 0. \quad (3.33)$$

For $m = 0$, it reduces to

$$i\gamma^\mu \partial_\mu \psi = 0, \quad (3.34)$$

which can be written explicitly as

$$i\gamma^0 \partial_t \psi + i\gamma^i \partial_i \psi = 0. \quad (3.35)$$

Multiplying from the left by γ^0 gives the Hamiltonian form

$$\gamma^0 (i\gamma^0 \partial_t \psi + i\gamma^i \partial_i \psi) = 0, \quad (3.36)$$

$$i\partial_t \psi + i\gamma^0 \gamma^i \partial_i \psi = 0. \quad (3.37)$$

Introducing the momentum operator $\vec{p} = -i\vec{\nabla}$ and defining

$$\alpha^i \equiv \gamma^0 \gamma^i, \quad (3.38)$$

we obtain the Schrödinger-like form

$$i\partial_t \psi = (\vec{\alpha} \cdot \vec{p})\psi. \quad (3.39)$$

—

Gamma matrix representation

We use the tensor-product representation

$$\gamma^0 = I \otimes \tau_x, \quad \gamma^i = \sigma_i \otimes i\tau_y, \quad (3.40)$$

where σ_i acts on spin space and τ_i acts on chiral (left/right) space.

The chirality operator is

$$\gamma^5 \equiv i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = -I \otimes \tau_z, \quad (3.41)$$

with eigenstates defined by

$$\gamma^5 \psi_\pm = \pm \psi_\pm \quad \Rightarrow \quad \tau_z \psi_\pm = \mp \psi_\pm. \quad (3.42)$$

—

Hamiltonian reduction

The Hamiltonian is

$$H = \gamma^0 (\vec{\gamma} \cdot \vec{p}) = \gamma^0 \gamma^i p_i. \quad (3.43)$$

Substituting the explicit representation,

$$H = (I \otimes \tau_x)(\sigma_i \otimes i\tau_y)p_i. \quad (3.44)$$

Using

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD),$$

$$H = \sigma_i \otimes (\tau_x i \tau_y) p_i. \quad (3.45)$$

Using Pauli matrix identities,

$$\tau_x \tau_y = i \tau_z \quad \Rightarrow \quad \tau_x (i \tau_y) = -\tau_z,$$

$$H = -\sigma_i \otimes \tau_z p_i = -(\vec{p} \cdot \vec{\sigma}) \otimes \tau_z. \quad (3.46)$$

—

Weyl decomposition

Projecting onto chiral eigenstates gives

$$H\psi_{\pm} = \mp(\vec{p} \cdot \vec{\sigma})\psi_{\pm}. \quad (3.47)$$

Thus the Dirac equation decouples into two Weyl equations

$$i\partial_t \psi_{\pm} = \mp(\vec{p} \cdot \vec{\sigma})\psi_{\pm}. \quad (3.48)$$

The corresponding Weyl Hamiltonians are

$$H_{\pm} = \mp \vec{p} \cdot \vec{\sigma}. \quad (3.49)$$

—

For plane-wave solutions $\psi \sim e^{-iEt + i\vec{p} \cdot \vec{x}}$, the dispersion relation is

$$E = \pm |\vec{p}|, \quad (3.50)$$

which describes relativistic massless Weyl fermions in $(3+1)$ dimensions. In this regime, the spin is locked parallel or antiparallel to momentum, defining the chirality of the particle.

Chapter 4

Some Basic Condensed Matter Physics

4.1 Bloch's Theorem

The electronic properties of a crystalline solid are governed by the single-particle Schrödinger equation

$$H\psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (4.1)$$

where the Hamiltonian is

$$H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{r}). \quad (4.2)$$

A crystal possesses a periodic lattice structure. Consequently, the potential experienced by an electron is periodic under translations by any lattice vector $\mathbf{R}$:

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r}). \quad (4.3)$$

Here $\mathbf{R}$ denotes an arbitrary lattice translation,

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \quad (4.4)$$

with $n_i \in \mathbb{Z}$ and $\mathbf{a}_i$ the primitive lattice vectors.

The periodicity of the potential implies that the Hamiltonian is invariant under lattice translations. To make this statement precise, we introduce the translation operator $T_{\mathbf{R}}$, defined through its action on a wavefunction:

$$T_{\mathbf{R}}\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{R}). \quad (4.5)$$

Since the potential remains unchanged under translations by lattice vectors, the Hamiltonian commutes with every translation operator,

$$[H, T_{\mathbf{R}}] = 0. \quad (4.6)$$

The commutation relation is of fundamental importance because commuting operators can be simultaneously diagonalized. Therefore, the energy eigenstates of the crystal may be chosen to be simultaneous eigenstates of the translation operator:

$$H\psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (4.7)$$

and

$$T_{\mathbf{R}}\psi(\mathbf{r}) = \lambda(\mathbf{R})\psi(\mathbf{r}), \quad (4.8)$$

where $\lambda(\mathbf{R})$ is the eigenvalue associated with the translation operator.

Using the definition of $T_{\mathbf{R}}$, the above equation becomes

$$\psi(\mathbf{r} + \mathbf{R}) = \lambda(\mathbf{R})\psi(\mathbf{r}). \quad (4.9)$$

Equation (4.9) states that translating the wavefunction by a lattice vector changes it only by a multiplicative factor.

To determine the form of $\lambda(\mathbf{R})$, we use the fact that lattice translations form a group. Performing two successive translations by $\mathbf{R}_1$ and $\mathbf{R}_2$ is equivalent to a single translation by $\mathbf{R}_1 + \mathbf{R}_2$:

$$T_{\mathbf{R}_1}T_{\mathbf{R}_2} = T_{\mathbf{R}_1+\mathbf{R}_2}. \quad (4.10)$$

Acting on the eigenstate ψ , the left-hand side gives

$$T_{\mathbf{R}_1}T_{\mathbf{R}_2}\psi = T_{\mathbf{R}_1}[\lambda(\mathbf{R}_2)\psi] \quad (4.11)$$

$$= \lambda(\mathbf{R}_2)T_{\mathbf{R}_1}\psi \quad (4.12)$$

$$= \lambda(\mathbf{R}_2)\lambda(\mathbf{R}_1)\psi. \quad (4.13)$$

The right-hand side gives

$$T_{\mathbf{R}_1+\mathbf{R}_2}\psi = \lambda(\mathbf{R}_1 + \mathbf{R}_2)\psi. \quad (4.14)$$

Equating both expressions, we obtain

$$\lambda(\mathbf{R}_1 + \mathbf{R}_2) = \lambda(\mathbf{R}_1)\lambda(\mathbf{R}_2). \quad (4.15)$$

Equation (4.15) is a multiplicative functional equation. The continuous solutions are exponential functions. Therefore, we may write

$$\lambda(\mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}}, \quad (4.16)$$

where $\mathbf{k}$ is a real vector known as the crystal momentum.

Substituting this result into Eq. (4.9), we find

$$\psi(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}}\psi(\mathbf{r}). \quad (4.17)$$

Equation (4.17) is known as the Bloch condition.

To isolate the non-periodic part of the wavefunction, we define a new function

$$u_{\mathbf{k}}(\mathbf{r}) = e^{-i\mathbf{k}\cdot\mathbf{r}}\psi_{\mathbf{k}}(\mathbf{r}). \quad (4.18)$$

We now examine how this function transforms under lattice translations:

$$u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{-i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})}\psi_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) \quad (4.19)$$

$$= e^{-i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})}e^{i\mathbf{k}\cdot\mathbf{R}}\psi_{\mathbf{k}}(\mathbf{r}) \quad (4.20)$$

$$= e^{-i\mathbf{k}\cdot\mathbf{r}}\psi_{\mathbf{k}}(\mathbf{r}) \quad (4.21)$$

$$= u_{\mathbf{k}}(\mathbf{r}). \quad (4.22)$$

Thus,

$$u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{\mathbf{k}}(\mathbf{r}), \quad (4.23)$$

which shows that $u_{\mathbf{k}}(\mathbf{r})$ has exactly the periodicity of the crystal lattice.

Finally, substituting the definition of $u_{\mathbf{k}}$ back into the wavefunction yields

$$\boxed{\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r})}, \quad (4.24)$$

with

$$u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r}). \quad (4.25)$$

This result is known as **Bloch's theorem**. It states that every eigenstate of an electron moving in a periodic crystal potential can be written as the product of a plane-wave factor $e^{i\mathbf{k}\cdot\mathbf{r}}$ and a lattice-periodic function $u_{n\mathbf{k}}(\mathbf{r})$. The index n labels the energy band, while $\mathbf{k}$ labels the crystal momentum within the Brillouin zone.

4.2 Energy Dispersion Relation - Free Electron

After understanding Bloch's theorem, we can immediately try to find the simplest energy dispersion relation. To keep the problem simple, we assume that $V(\mathbf{r}) = 0$ and that the state ψ is constant, without worrying about how closely this resembles reality. As a practice, we will see how the Schrödinger equation and the Bloch state work together perfectly.

As usual, we start with the time-independent Schrödinger equation,

$$\left[\frac{\hat{p}^2}{2m} + V(\mathbf{r}) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}). \quad (4.26)$$

The Bloch state is given by

$$\psi_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (4.27)$$

Substituting Eq. (4.27) into Eq. (4.26), we obtain

$$\left[\frac{\hat{p}^2}{2m} + V(\mathbf{r}) \right] u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}} = E u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (4.28)$$

Since we are considering the free-electron model, we set

$$V(\mathbf{r}) = 0. \quad (4.29)$$

The next step is to understand how the momentum operator acts on the Bloch state. Using

$$\hat{p} = -i\hbar\nabla, \quad (4.30)$$

we obtain

$$\begin{aligned} \hat{p} [u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}] &= -i\hbar\nabla [u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}] \\ &= e^{i\mathbf{k}\cdot\mathbf{r}} (-i\hbar\nabla + \hbar\mathbf{k}) u_{n\mathbf{k}}(\mathbf{r}) \\ &= e^{i\mathbf{k}\cdot\mathbf{r}} (\hat{p} + \hbar\mathbf{k}) u_{n\mathbf{k}}(\mathbf{r}). \end{aligned} \quad (4.31)$$

Thus, inside the Bloch state, the momentum operator effectively transforms as

$$\hat{p} \rightarrow \hat{p} + \hbar\mathbf{k}. \quad (4.32)$$

Substituting this result into Eq. (4.28) gives

$$\frac{(\hat{p} + \hbar\mathbf{k})^2}{2m} u_{n\mathbf{k}}(\mathbf{r}) = E u_{n\mathbf{k}}(\mathbf{r}). \quad (4.33)$$

For the simplest free-electron case, we take $u_{n\mathbf{k}}(\mathbf{r})$ to be a constant. Therefore,

$$\hat{p} u_{n\mathbf{k}}(\mathbf{r}) = 0. \quad (4.34)$$

Cancelling $u_{n\mathbf{k}}(\mathbf{r})$ from both sides, we obtain

$$E = \frac{(\hbar\mathbf{k})^2}{2m}. \quad (4.35)$$

Hence, the energy dispersion relation for a free electron is

$$E(\mathbf{k}) = \frac{\hbar^2 k^2}{2m}. \quad (4.36)$$

4.3 Periodic Potential

Periodic Potential

We will take things one step further. From now on, the potential $V(\mathbf{r})$ is no longer zero. However, we are not going to consider the most general case. In fact, one may wonder whether the Schrödinger equation can be solved at all without any underlying symmetries.

The additional information we now have is that the potential is periodic. Therefore,

$$V(x + L) = V(x), \quad (4.37)$$

where L is the lattice periodicity.

From Bloch's theorem, we know that the wavefunction must satisfy

$$\psi_k(x + L) = e^{ikL}\psi_k(x), \quad (4.38)$$

or equivalently,

$$\psi_k(x) = u_k(x)e^{ikx}, \quad (4.39)$$

where the function $u_k(x)$ has the same periodicity as the lattice,

$$u_k(x + L) = u_k(x). \quad (4.40)$$

Substituting Eq. (4.39) into the time-independent Schrödinger equation,

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi_k(x) = E\psi_k(x), \quad (4.41)$$

we obtain a differential equation with periodic coefficients.

Our task is now to solve this particular differential equation.

Fourier Modes

The next step is to find a way to reduce the complexity of the problem and transform it into something much simpler. Fortunately, there exists a powerful mathematical theorem known as the Fourier theorem. It states that any periodic function can be represented as a superposition of simple oscillatory functions such as sines and cosines. Equivalently, using Euler's identity, we may work with complex exponentials.

Since the potential $V(x)$ is periodic, it is natural to expand it in terms of Fourier modes,

$$V(x) = \sum_G V_G e^{iGx}, \quad (4.42)$$

where V_G are the Fourier coefficients.

At this stage, the quantity G is unknown. However, because the potential satisfies

$$V(x + L) = V(x), \quad (4.43)$$

each Fourier mode must individually satisfy the same periodicity condition. Therefore,

$$e^{iG(x+L)} = e^{iGx}. \quad (4.44)$$

Dividing both sides by e^{iGx} , we obtain

$$e^{iGL} = 1. \quad (4.45)$$

Using Euler's identity,

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (4.46)$$

the above condition is satisfied only when

$$GL = 2\pi n, \quad n \in \mathbb{Z}. \quad (4.47)$$

Hence,

$$G = \frac{2\pi n}{L}, \quad n = 0, \pm 1, \pm 2, \dots \quad (4.48)$$

and the Fourier expansion of the periodic potential becomes

$$V(x) = \sum_n V_n e^{i \frac{2\pi n}{L} x}. \quad (4.49)$$

It is important to remember that the quantity G will appear repeatedly in our future discussions. It is not merely a mathematical trick introduced to simplify calculations. Rather, it reflects a physical property of the underlying periodic structure. As we proceed to higher dimensions, these quantities will acquire a deeper geometrical meaning and will form the foundation of what is known as the reciprocal lattice.

An Example: Solving a Differential Equation Using Fourier Modes

Before returning to the Schrödinger equation, let us consider a simple example to understand why Fourier modes are so useful. Consider the differential equation

$$\frac{d^2 y}{dx^2} + 4y = \cos(3x), \quad (4.50)$$

subject to the periodic boundary condition

$$y(x + 2\pi) = y(x). \quad (4.51)$$

At first sight, this appears to be a problem involving calculus. However, since the solution is periodic, it is natural to expand it in Fourier modes,

$$y(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}. \quad (4.52)$$

Taking derivatives is now straightforward. Differentiating once gives

$$\frac{dy}{dx} = \sum_n in c_n e^{inx}, \quad (4.53)$$

and differentiating a second time gives

$$\frac{d^2 y}{dx^2} = \sum_n (-n^2) c_n e^{inx}. \quad (4.54)$$

Notice what has happened. The differential operator

$$\frac{d^2}{dx^2} \quad (4.55)$$

has become multiplication by

$$-n^2. \quad (4.56)$$

This is the main advantage of working in the Fourier basis.

Using Euler's identity,

$$\cos(3x) = \frac{1}{2} (e^{i3x} + e^{-i3x}), \quad (4.57)$$

and substituting Eqs. (4.52) and (4.54) into Eq. (4.50), we obtain

$$\sum_n (4 - n^2) c_n e^{inx} = \frac{1}{2} (e^{i3x} + e^{-i3x}). \quad (4.58)$$

Since the Fourier modes are linearly independent, the coefficients of each mode must match. Therefore,

$$(4 - n^2) c_n = 0, \quad n \neq \pm 3, \quad (4.59)$$

while for $n = \pm 3$,

$$(4 - 9) c_{\pm 3} = \frac{1}{2}. \quad (4.60)$$

Thus,

$$c_{\pm 3} = -\frac{1}{10}, \quad (4.61)$$

and all other coefficients vanish.

The solution is therefore

$$\begin{aligned} y(x) &= -\frac{1}{10} e^{i3x} - \frac{1}{10} e^{-i3x} \\ &= -\frac{1}{5} \cos(3x). \end{aligned} \quad (4.62)$$

The important lesson is not the solution itself, but the method. A differential equation involving derivatives has been converted into a set of algebraic equations for the Fourier coefficients. In this sense, Fourier modes transform a calculus problem into a linear algebra problem. This is precisely the strategy that we will employ when solving the Schrödinger equation in a periodic potential.

Solving the Schrödinger Equation

As discussed earlier, our goal is now to solve the time-independent Schrödinger equation

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi_k(x) = E \psi_k(x). \quad (4.63)$$

Rather than working directly with the differential equation, we shall exploit the periodicity of the system and work in a Fourier basis.

From Bloch's theorem, the wavefunction can be written as

$$\psi_k(x) = u_k(x)e^{ikx}, \quad (4.64)$$

where $u_k(x)$ has the same periodicity as the lattice,

$$u_k(x + L) = u_k(x). \quad (4.65)$$

Since $u_k(x)$ is periodic, it can be expanded in Fourier modes,

$$u_k(x) = \sum_G c_G e^{iGx}. \quad (4.66)$$

Substituting this into Bloch's form gives

$$\psi_k(x) = \sum_G c_G e^{i(k+G)x}. \quad (4.67)$$

Similarly, the periodic potential can be expanded as

$$V(x) = \sum_{G'} V_{G'} e^{iG'x}. \quad (4.68)$$

Substituting both expansions into the Schrödinger equation yields

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \sum_{G'} V_{G'} e^{iG'x} \right] \sum_G c_G e^{i(k+G)x} = E \sum_G c_G e^{i(k+G)x}. \quad (4.69)$$

At first sight this expression appears more complicated than the original differential equation. However, a remarkable simplification occurs because the Fourier modes are eigenfunctions of the derivative operator.

Acting with the kinetic-energy operator gives

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} e^{i(k+G)x} = \frac{\hbar^2(k+G)^2}{2m} e^{i(k+G)x}. \quad (4.70)$$

Thus the differential operator has been converted into multiplication by a number. This is precisely the advantage of working in the Fourier basis.

The Schrödinger equation therefore becomes

$$\sum_G \frac{\hbar^2(k+G)^2}{2m} c_G e^{i(k+G)x} + \sum_{G,G'} V_{G'} c_G e^{i(k+G+G')x} = E \sum_G c_G e^{i(k+G)x}. \quad (4.71)$$

Our next task will be to collect terms with identical Fourier factors and derive a set of algebraic equations for the coefficients c_G . This will transform the original differential equation into a matrix eigenvalue problem, known as the central equation. We now focus on the potential term,

$$\sum_{G,G'} V_{G'} c_G e^{i(k+G+G')x}. \quad (4.72)$$

The difficulty is that the exponential contains the combination $G + G'$, whereas the kinetic-energy and right-hand-side terms are written in terms of the basis functions $e^{i(k+G)x}$. In order to

compare coefficients of identical Fourier modes, we must rewrite the potential term in the same basis.

To achieve this, we introduce a new reciprocal lattice vector

$$G'' = G + G'. \quad (4.73)$$

Equivalently,

$$G' = G'' - G. \quad (4.74)$$

Substituting this into the potential term gives

$$\sum_{G,G'} V_{G'} c_G e^{i(k+G+G')x} = \sum_{G,G''} V_{G''-G} c_G e^{i(k+G'')x}. \quad (4.75)$$

Notice that the exponential now has the desired form $e^{i(k+G'')x}$.

Since G and G'' are independent summation indices, we may interchange the order of summation,

$$\sum_{G,G''} V_{G''-G} c_G e^{i(k+G'')x} = \sum_{G''} \left(\sum_G V_{G''-G} c_G \right) e^{i(k+G'')x}. \quad (4.76)$$

The Schrödinger equation therefore becomes

$$\sum_G \frac{\hbar^2(k+G)^2}{2m} c_G e^{i(k+G)x} + \sum_{G''} \left(\sum_G V_{G''-G} c_G \right) e^{i(k+G'')x} = E \sum_G c_G e^{i(k+G)x}. \quad (4.77)$$

The symbol used for a summation index is arbitrary. Therefore we may relabel

$$G'' \rightarrow G, \quad (4.78)$$

which yields

$$\sum_G \frac{\hbar^2(k+G)^2}{2m} c_G e^{i(k+G)x} + \sum_G \left(\sum_{G'} V_{G-G'} c_{G'} \right) e^{i(k+G)x} = E \sum_G c_G e^{i(k+G)x}. \quad (4.79)$$

All terms are now expressed in the common Fourier basis $e^{i(k+G)x}$. We may therefore combine them into a single summation,

$$\sum_G \left[\frac{\hbar^2(k+G)^2}{2m} c_G + \sum_{G'} V_{G-G'} c_{G'} - E c_G \right] e^{i(k+G)x} = 0. \quad (4.80)$$

Since the Fourier modes $e^{i(k+G)x}$ are linearly independent, the coefficient multiplying each mode must vanish separately. Consequently,

$$\left[\frac{\hbar^2(k+G)^2}{2m} - E \right] c_G + \sum_{G'} V_{G-G'} c_{G'} = 0. \quad (4.81)$$

This is the central equation.

Some notes on reciprocal lattcie

Conclusion